

SVILENGRAD, Bulgaria

WMO: 157410

Lat: 41.767N

Lon: 26.200E

Elev: 55

StdP: 100.67

Time Zone: 2.00 (EUE)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.2	-11.2	1.4	-7.4	-9.0	1.8	-4.5	9.8	5.5	8.1	3.9	0.8	N/A	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	35.4	22.2	33.8	21.7	32.3	21.0	23.5	33.2	22.4	31.8	21.7	30.7	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.8	28.2	19.2	14.1	26.6	18.5	13.5	25.8	70.1	33.5	66.1	32.0	63.3	30.7	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.7	5.6	-13.5	39.1	3.9	2.2	-16.4	40.7	-18.6	41.9	-20.8	43.1	-23.7	44.7		
(5)			-13.0	25.3	3.1	0.7	-15.2	25.8	-17.0	26.3	-18.8	26.7	-21.0	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	2.5	4.8	8.0	12.5	18.0	22.5	25.2	24.5	19.9	14.3	7.9	2.7		
	DBStd	8.89	4.01	4.30	4.12	3.85	3.18	2.73	2.36	2.55	3.30	4.10	4.19	5.35		
	HDD10.0	827	236	149	92	21	1	0	0	0	0	9	88	232		
	HDD18.3	2371	492	378	321	175	44	3	1	0	22	138	313	483		
	CDD10.0	2147	2	4	29	97	248	374	472	450	297	142	25	7		
	CDD18.3	649	0	0	0	1	33	128	214	192	69	12	1	0		
	CDH23.3	7166	0	0	6	19	424	1407	2437	2119	655	98	2	0		
	CDH26.7	3118	0	0	1	1	117	566	1196	1012	202	23	0	0		
(14) Wind	WSAvg	1.6	1.4	1.8	2.0	1.9	1.5	1.6	1.8	1.5	1.4	1.3	1.2	1.4		
(15) Precipitation	PrecAvg	556	39	40	42	46	55	37	43	31	43	52	62	66		
	PrecMax	861	123	125	108	80	156	106	165	56	136	106	118	138		
	PrecMin	440	2	7	5	11	15	3	1	0	0	4	15	31		
	PrecStd	112	34	30	27	20	39	26	38	16	39	32	32	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	18.5	23.8	25.7	32.4	35.3	38.9	37.8	33.4	29.4	22.0	17.4		
		MCWB	11.6	12.2	15.2	16.1	20.7	21.6	22.3	22.8	20.2	18.9	16.1	13.4		
	2%	DB	12.5	16.1	19.7	23.7	29.5	33.1	35.7	35.3	31.1	25.8	19.1	15.3		
		MCWB	9.6	10.9	13.1	15.3	19.0	21.5	22.6	22.1	19.6	18.2	14.4	12.4		
	5%	DB	10.0	14.4	17.7	21.7	27.4	31.5	34.0	33.7	29.2	23.4	16.8	12.9		
		MCWB	7.9	9.9	12.0	14.5	18.3	21.0	22.0	21.6	19.1	17.1	13.5	10.7		
	10%	DB	7.9	12.1	15.6	19.8	25.5	30.0	32.4	32.0	27.2	21.4	14.5	10.5		
		MCWB	6.6	8.6	10.6	13.6	17.3	20.1	21.2	20.8	18.3	15.7	12.1	9.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	12.6	15.5	17.1	21.8	23.3	24.8	24.8	21.2	19.7	16.2	13.7		
		MCDB	15.2	17.4	22.1	23.7	30.5	32.4	34.9	35.5	29.3	27.1	20.6	16.8		
	2%	WB	10.1	11.4	13.7	15.9	19.8	22.3	23.7	23.3	20.2	18.7	15.0	12.6		
		MCDB	12.4	15.6	19.2	21.8	27.6	31.3	33.8	33.1	28.4	24.8	18.1	15.1		
	5%	WB	8.2	10.1	12.3	15.0	18.8	21.5	22.6	22.2	19.6	17.8	13.8	11.0		
		MCDB	9.5	13.9	17.1	20.4	26.2	30.2	32.0	31.8	27.5	22.9	16.9	12.4		
	10%	WB	6.6	8.8	11.1	14.2	17.9	20.7	21.8	21.3	18.8	16.5	12.1	9.0		
		MCDB	7.8	12.0	15.0	19.0	24.6	28.8	31.0	30.4	26.1	20.6	14.2	10.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	10.1	10.8	10.9	13.2	13.7	13.6	13.7	12.5	10.4	8.4	6.7		
		MCDBR	10.6	14.3	14.1	14.4	17.0	16.5	16.0	16.4	15.3	13.7	12.3	9.0		
		MCWBR	7.9	9.4	8.2	7.2	8.1	6.9	5.8	6.2	6.1	6.7	7.9	6.5		
	5% WB	MCDBR	9.5	13.5	13.1	12.3	15.4	15.4	14.8	14.9	13.6	12.3	10.9	8.1		
		MCWBR	7.6	9.0	7.8	6.6	7.8	7.1	6.1	6.1	5.8	6.4	7.5	6.2		
(40) Clear-Sky Solar Irradiance	taub	0.349	0.380	0.422	0.472	0.449	0.449	0.449	0.449	0.456	0.436	0.404	0.375	0.355		
	taud	2.336	2.245	2.143	2.015	2.161	2.198	2.211	2.187	2.212	2.284	2.315	2.313			
	Ebn at Noon	781	810	815	800	832	830	826	807	793	773	740	733			
	Edh at Noon	87	110	137	167	148	143	140	139	127	105	88	82			
(44) All-Sky Solar Radiation	RadAvg	1.60	2.40	3.52	4.80	6.01	6.55	6.82	6.14	4.55	2.96	1.86	1.41			
	RadStd	0.20	0.26	0.31	0.51	0.41	0.51	0.41	0.28	0.32	0.29	0.20	0.20			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.47	N/A	N/A	N/A	+0.58	+0.62	-89	-129	+92	N/A

Nomenclature: See separate page